MEAL-163 | Development Prerequisites & Reproducible Local Setup

MEAL CONNECT • TEAM APEX • VERSION 1.3 • 01 NOV 2026

Purpose: provide one repeatable Windows localhost setup for all seven developers. This guide records required runtimes, compatibility assumptions, project commands, configuration boundaries and a verification procedure. It describes the intended development baseline, not a claim that every service is already implemented.

1. Confirmed project baseline

Area	Chosen approach
Frontend	Angular 21 SPA, TypeScript, SCSS and Angular Router
Backend	Python / FastAPI REST API
Database	PostgreSQL
Runtime	Localhost; Docker not required
Repository	meal-connect; frontend/, backend/, database/, docs/; feature branches → PR → develop
MVP business rule	NGO requests quantity from an existing listing; restaurant approves whole request or rejects; prevent over-allocation

2. Required software and version policy

Tool	Requirement / baseline	Verification
Node.js	Angular 21 compatible: 20.19+, 22.12+, or 24.0+ within those supported majors (prefer Node 22 LTS for team consistency)	node --version
npm	Version bundled with the chosen Node distribution; use repository lockfile	npm --version
Angular / CLI	21.x; local package.json and package-lock.json are authoritative	npx ng version
TypeScript	>= 5.9.0 and < 6.0.0 for Angular 21; resolved version is locked by npm	npm ls typescript
RxJS	^6.5.3 or ^7.4.0 per Angular 21 compatibility	npm ls rxjs
Python	Suggested project baseline: Python 3.12; freeze a single minor version across team after verification	py -3.12 --version
FastAPI / Uvicorn	Pin tested versions in backend/requirements.txt; not yet established by repository	python -m pip show fastapi uvicorn

PostgreSQL	Suggested project baseline: PostgreSQL 16; confirm installed major and migrations compatibility	psql --version
Git	Git CLI available; use feature branch workflow	git --version

Important: Python 3.12 and PostgreSQL 16 are reproducibility recommendations, not confirmed installed versions for every member. Verify before enforcing. Do not install Angular 22 merely because it is newer. Official Angular 21 compatibility: https://angular.dev/reference/versions

3. Verify a developer machine (PowerShell)

```
git --version
node --version
npm --version
cd frontend
npx ng version
npm ls typescript rxjs
cd ..
py --list-paths
py -3.12 --version
psql --version
```

If py -3.12 is unavailable, record the installed Python minor version; select one agreed version before setting up backend dependencies. PostgreSQL CLI may require its bin directory on PATH. The above checks must be run on each developer machine, not assumed from installation alone.

4. Reproduce the frontend (existing Angular project)

```
git clone https://github.com/rivipahanviksura/meal-connect.git
cd meal-connect
git switch develop
cd frontend
npm ci
npm start
```

Open http://localhost:4200/ and confirm the Angular page loads. Use npm ci, not npm install, for reproducible installs from frontend/package-lock.json. Never commit node_modules/, dist/ or .angular/cache/. Run commands from frontend/ when instructed.

5. Prepare a local Python backend environment

FastAPI code and the dependency lock list may be created in a later development task. Until those files exist, the following is the agreed setup pattern; do not report the API as running merely because Python is installed.

```
cd backend
py -3.12 -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
# AFTER backend/requirements.txt has been committed:
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

```
# AFTER backend/app/main.py exposes FastAPI app:
.\.venv\Scripts\python.exe -m uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

Preferred backend layout: backend/app/main.py, api/, core/, models/, schemas/, services/, db/ and tests/. Treat this layout as a target convention; do not create placeholder application behavior to satisfy a checklist.

6. PostgreSQL service and local configuration

Install/start the local PostgreSQL server. Provision one development-only database and a least-privilege local role; database creation requires suitable local PostgreSQL administrator access. Never embed actual database passwords in documentation or source code.

```
psql -h localhost -U postgres -c "SELECT version();"
# Execute DB creation / user grants using your local SQL administration tool.
# Apply versioned SQL migrations when the repository includes them.
```

The logical model is defined by MEAL-6: 10 tables, relational constraints, 0..1 pickup per request, 0..1 distribution per pickup, and transaction-safe quantity allocation. DBML is not itself an executable migration.

7. Environment variables and secret hygiene

Variable	Example / purpose
APP_ENV	development
APP_HOST / APP_PORT	127.0.0.1 / 8000
DATABASE_URL	<local PostgreSQL connection URL; set privately>
JWT_SECRET_KEY	<unique random value generated per developer>
JWT_ALGORITHM	HS256 (MEAL-62 baseline)
ACCESS_TOKEN_EXPIRE_MINUTES	30 (MEAL-63 example)
CORS_ORIGINS	http://localhost:4200
LOG_LEVEL	INFO

Put real settings in backend/.env (untracked). Commit only backend/.env.example containing placeholders and non-sensitive defaults. Add .env and .venv to .gitignore and check git status before every push. A browser-side Angular environment file must NEVER contain signing keys or database credentials.

8. Local integration verification

Check	Expected result
Frontend	npm ci succeeds and http://localhost:4200/ renders
Backend (once implemented)	Uvicorn starts and http://127.0.0.1:8000/docs is reachable
Database	Local PostgreSQL accepts authorized test connection
CORS	Angular origin http://localhost:4200 is allowed by FastAPI
API contract	Backend routes/status/error envelopes match MEAL-62

Security	No .env, credentials, .venv, node_modules, or build output staged in Git
Workflow	Donation approval allocation is atomic and cannot oversubscribe

9. Team version record (complete on each laptop)

Developer	Node / npm	Angular / TypeScript	Python / FastAPI	PostgreSQL
Member 1	To record	To record	To record	To record
Member 2	To record	To record	To record	To record
Member 3	To record	To record	To record	To record
Member 4	To record	To record	To record	To record
Member 5	To record	To record	To record	To record
Member 6	To record	To record	To record	To record
Member 7	To record	To record	To record	To record

Record outputs and date in the team setup log; do not claim “all identical” until versions match. If versions differ, use the supported Angular ranges and agree a common runtime before troubleshooting. Re-run npm ci after checking out the latest develop branch.

10. Document cross-references

MEAL-5: system architecture; MEAL-61: component data flow; MEAL-6: PostgreSQL model; MEAL-62: API endpoint contracts; MEAL-63: environment variables, CORS, JWT and backend configuration. These guides define the expected implementation. The actual source code and pinned dependencies are the executable reference once implemented.